Renke Wang

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Professional Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in mathematical optimization, resource-constrained scheduling, resource allocation, routing and path planning, and simulation for networked and multi-agent systems. Formulates mixed-integer and nonlinear optimization models and develops centralized, distributed, and tree/graph-search methods for operational planning, task sequencing, route and path selection, and real-time scheduling. Implements Python and MATLAB optimization and event-driven simulation workflows using Gurobi and CasADi, with experience in computational benchmarking, sensitivity analysis, and runtime and scalability evaluation for warehouse-robot coordination, automated-vehicle scheduling, and networked cyber-physical systems.

Education

George Mason University *Expected 2027*
Ph.D. Candidate in Electrical and Computer Engineering
Dalian University of Technology
Master of Engineering, Control Science and Engineering
Dalian University of Technology
Bachelor of Engineering, Automation

Technical Skills

Operations Research and Mathematical Optimization: Formulation and solution of Linear Programming (LP), Mixed-Integer Programming (MIP), and Nonlinear Programming (NLP) problems, including Mixed-Integer Linear Programming (MILP), Mixed-Integer Nonlinear Programming (MINLP), and nonconvex constrained formulations; centralized optimization and distributed consensus optimization using the Alternating Direction Method of Multipliers (ADMM), including the Consensus Alternating Direction Method of Multipliers (C-ADMM).

Scheduling, Planning, and Decision-Making: Resource-constrained, real-time, and priority-based scheduling, including priority assignment and dispatching; job-shop, flow-shop, and hybrid-shop scheduling structures; resource allocation, task allocation, and resource-occupation sequencing; graph-based routing and shortest-path planning; joint path planning and scheduling; online decision-making and multi-agent coordination under shared-resource constraints.

Optimal Control and Real-Time Systems: State-space modeling and feedback controller design for constrained dynamical systems; optimal control and receding-horizon optimization; Model Predictive Control (MPC), including robust and tube-based MPC; scheduling and control co-design for real-time systems; Linear Matrix Inequality (LMI)-based controller synthesis and Lyapunov-based stability analysis for positive systems, switched systems, and Markov Jump Linear Systems (MJLS); event-triggered control and robust control under timing uncertainty; Cyber-Physical Systems (CPS) and Networked Control Systems (NCS).

Algorithms, Simulation, and Learning: Algorithm design, decision-tree search, and graph-search methods, including Dijkstra's algorithm and A* search; multi-step lookahead; event-driven and Monte Carlo simulation; numerical optimization workflows; scenario and sensitivity analysis; model validation and benchmarking; solver diagnostics; runtime, computational performance, and scalability evaluation; supervised neural policy learning from exact tree-search decisions; deep reinforcement learning (DRL), including Proximal Policy Optimization (PPO).

Programming and Scientific Computing: Python (NumPy, SciPy, and Matplotlib); MATLAB; Gurobi; CasADi; YALMIP; MATLAB Optimization Toolbox (fmincon); modular optimization and simulation workflows; numerical debugging; reproducible computational experiments; Git and GitHub; LaTeX.

Research Experience

Research Assistant at George Mason University *Fairfax, VA*
August 2022 – Present

Optimization and Simulation for Network-Constrained Spatial Resource Scheduling

- Formulated and applied Contention-Resolving MPC to optimize operational timing and shared-resource allocation for warehouse robots and automated vehicles following predefined routes; coordinated resource-access sequencing and occupation timing across warehouse merging zones, multi-lane intersections, and multi-intersection traffic networks.
- Built event-driven browser-based simulations and animations to visualize robot and vehicle trajectories, resource-occupation timelines, scheduling decisions, and cost evolution.

Distributed Scheduling and Control Co-Design for NCS with Multi-Layered Shared Resources

- Developed and implemented a C-ADMM-based distributed optimization framework that decomposes cross-layer scheduling, sequencing, and resource-allocation decisions into resource-specific subproblems, enabling parallel execution of local optimization problems; evaluated system performance, computation time, and scalability against centralized optimization and established priority-based scheduling methods.

Scheduling and Control Co-Design under Heterogeneous Resources

- Developed and evaluated a centralized Contention-Resolving MPC framework for scheduling and control co-design under heterogeneous resource constraints, with Monte Carlo performance evaluation for multiple lighter-than-air agents (blimps) and additional applications to human–robot collaboration and game-theoretic multi-robot obstacle avoidance.

Learning-Based Path Selection and Contention-Resolving Scheduling

- Implemented a supervised neural scheduling policy trained on optimal scheduling decisions generated by exact Contention-Resolving MPC decision-tree search for event-driven contention resolution under shared-resource constraints.
- Developing an event-driven PPO actor-critic framework for joint path selection and shared-resource scheduling, extending fixed-route scheduling to online path-selection and priority decisions for multi-agent systems.

Robust Control of NCS under Distribution-Free Timing Perturbations

- Developed a tube-based robust MPC framework for NCS subject to distribution-free timing perturbations; designed dynamic controller gains and established stability guarantees for the resulting closed-loop error dynamics.

Graduate Researcher at Dalian University of Technology *Dalian, China*
Sept 2017 – June 2020

Stability Analysis and Controller Design for Positive and Switched Systems

- Derived Lyapunov-based stability conditions for positive switched systems and MJLS with dwell-time constraints, stochastic mode transitions, and fully or partially known transition rates.
- Designed state-feedback and L1 controllers using piecewise-linear copositive Lyapunov functions, establishing stochastic and exponential stability and L1-gain performance under external disturbances and controller failures.

Selected Publications and Manuscripts

- Renke Wang and Ningshi Yao. *Distributed Scheduling and Control Co-Design for Networked Control Systems with Multi-Layered Shared Resources.* IEEE Transactions on Control of Network Systems (TCNS), under review.
- Renke Wang, Mengxue Hou, and Ningshi Yao. *Robust Co-Design of Scheduling and Control for Lighter-than-Air Agents with Heterogeneous Resources.* International Journal of Robust and Nonlinear Control, under review.
- J. Lian and R. Wang. *Stochastic Stability of Positive Markov Jump Linear Systems with Fixed Dwell Time.* Nonlinear Analysis: Hybrid Systems, 40 (2021): 101014.

- R. Wang and N. Yao. *Contention-Resolving Model Predictive Control for Coordinating Automated Vehicles at a Multi-Lane Traffic Intersection.* 2025 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2025.
- Rajul Kumar, Renke Wang, and Ningshi Yao. *One Human and Multi-Robot Collaboration: Evaluating the Impact of Attention-Guided On-Screen Recommendations on Human Multitasking Efficiency, Cognitive Load, and Trust.* 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), IEEE, 2025.
- R. Wang and N. Yao. *Robust Model Predictive Control for Networked Control Systems with Timing Perturbations.* 2024 American Control Conference (ACC). IEEE, 2024.
- N. Yao and R. Wang. *Optimal Real-Time Human Attention Allocation and Scheduling in a Multi-human and Multi-robot Collaborative System.* 2024 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2024.

Teaching Experience

Teaching Assistant, George Mason University *Fairfax, VA*
Aug 2022 – Dec 2022

- Delivered lectures and led or co-led laboratory sessions for Continuous-Time Signals and Systems; explained technical concepts, guided laboratory work, graded assignments, and supported undergraduate students.
- Graded coursework and answered student questions for Classical Systems and Control Theory, providing clarification on course concepts and assignments.

Honors and Professional Service

- Summer Research Assistantship Award, George Mason University Graduate Division, 2026.
- Reviewer for IEEE Transactions on Control Systems Technology; IEEE Transactions on Systems, Man, and Cybernetics: Systems; and IEEE conferences including ACC, CDC, and CCTA, 2023–Present.